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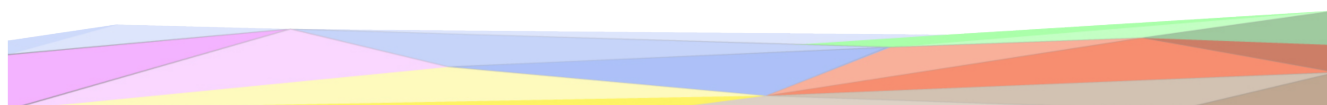
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Index:

1	Introduction	1
2	Project Design	1
3	Purpose Definition	2
3.1	Purpose Formalization	2
3.1.1	Scenarios Definition	2
3.1.2	Personas	2
3.1.3	Competency Questions (CQs)	3
3.1.4	Concepts Identification	3
3.1.5	ER Model	5
3.2	Information Gathering	6
3.2.1	Knowledge Layer	6
3.2.2	Data Layer	6
3.3	Report	9
3.3.1	Personas and Scenarios	9
3.3.2	Competency Questions (CQs)	9
3.3.3	Entity Relationship Diagrams	10
4	Language Definition	11
4.1	Knowledge Layer	11
5	Knowledge Definition	14
6	Entity Definition	15
7	Evaluation	16
8	Metadata Definition	17
9	Open Issues	17

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1 Introduction

Reusability is one of the main principles in the Knowledge Graph (KG) development process defined by iTelos. The KG project documentation plays an important role to enhance the reusability of the resources handled and produced during the process. A clear description of the resources, the process (and sub processes) developed and evaluation at each step of the process provides a clear understanding of the project, thus serving such an information to external readers for the future exploitations of the project's outcomes.

The current document aims to provide a detailed report of the project developed following the iTelos methodology. The report is structured, to describe:

- Section 2: Definition of the project's purpose and related information gathering.
- Sections 3, 4, 5, 6: The description of the iTelos process phases and their activities, divided by knowledge and data layer activities, as well as the evaluation of the resources produced in terms of fit for the chosen purpose.
- Section 7: The description of the metadata produced for all (and all kind of) the resources handled and generated by the iTelos process, while executing the project.
- Section 8: Conclusion and open issues summary.

2 Project Design

The primary goal of this project is to build a **Knowledge Graph (KG)** that stores, organizes, and provides easy access to information about **essential geographical features and various healthcare facilities** within a specific territory, namely the **Trentino region**. The KG will function as a centralized knowledge hub, connecting citizens, public health planners, and emergency services to critical information regarding the **location, specialization, availability, and accessibility** of healthcare resources across the region.

The **Domain of Interest (DoI)** for the "Territory and Health Facilities" project is the **Trentino region**, focusing specifically on the relationship between its **geographical features** and **healthcare infrastructure**.

The DoI encompasses the following key aspects:

- **Geographical Scope:** The Trentino region (e.g., municipalities, towns, and locations) is the territory under study.
- **Target Entities:** Essential geographical features and different healthcare facilities (hospitals, clinics, specialized centers, emergency medical facilities) within that territory.
- **Project Focus:** Modeling the location, specialization, availability, and, crucially, the accessibility/proximity of these facilities relative to geographical points (towns, given locations).

3 Purpose Definition

3.1 Purpose Formalization

3.1.1 Scenarios Definition

In this section, a set of usage scenarios, describing the multiple aspects considered by the project purpose (linking geographical features to healthcare infrastructure), are presented.

1. **Emergency Proximity** A tourist, Marco, is hiking in a remote area (he only knows his GPS coordinates). He has a minor emergency and needs to find the absolute nearest hospital and pharmacy, regardless of the municipality.
2. **Local Specialized Care Access** A caregiver, Giulia, lives in the municipality of 'Cles'. She needs to find a semi-residential facility for her elderly father. She wants to find all options within her municipality first, and then expand her search to nearby municipalities if necessary.
3. **Service Gap & Distribution Analysis** A public health analyst, Dr. Rossi, is tasked with identifying potential "service deserts." She needs to map the locations of all psychiatric services to see which municipalities have no local access.
4. **Territorial Capacity Planning** A public health planner, Luca, is reviewing the total capacity for long-term care. He needs to know the total number of beds available in residential facilities for the elderly, aggregated by geographical area (e.g., all facilities in the municipality of 'Trento').

3.1.2 Personas

1. **Marco Bianchi** is a 32-year-old tourist visiting Trentino for a hiking trip. His main concern is safety in unfamiliar territory. He needs to be able to find the nearest hospital or pharmacy based on his phone's GPS coordinates in case of an emergency, rather than searching by a town name he doesn't know.
2. **Giulia Verdi** is a 52-year-old resident of Cles. Her main task is finding a suitable care facility for her elderly father. She needs to find all residential or semi-residential facilities that specifically offer Elderly Assistance within her home municipality.
3. **Dr. Elena Rossi** is a 45-year-old public health analyst. Her main task is creating reports on healthcare access and equity. She needs to identify which municipalities do not have a local facility offering specialized services like Psychiatric Assistance to map service deserts.
4. **Luca Verdi** is a 38-year-old health system planner. His main task is managing resource allocation for the upcoming year. He needs to know the total number of beds available in residential facilities aggregated by municipality to assess regional capacity.

3.1.3 Competency Questions (CQs)

Initially, we came up with some scratch competency questions that were not based on actual data, but an imaginary idea of what the datasets and data could include. After finalizing the datasets chosen for this project, the questions were either modified or deleted to fit the datasets better, this included an improved set of competency questions that are more precise and helpful. (make more formal, questions include emergency details that got terminated later, usage of ontological aspects)

Table 1: *Competency Questions (CQs)*

Person	No.	Question
Marco Bianchi	1.1	What is the nearest hospital to a current GPS coordinates?
Marco Bianchi	1.2	Find the address and phone number of a specific health structure.
Marco Bianchi	1.3	List all pharmacies within a 5km radius of a specific health facility.
Giulia Verdi	2.1	Which residential facilities for Elderly Assistance are located in a specific municipality?
Giulia Verdi	2.2	Are there any semi-residential structures in a specific municipality? If not, what are the closest ones?
Giulia Verdi	2.3	What specialized services are available at the health structures in a specific municipality?
Dr. Elena Rossi	3.1	Which municipalities do not have any health structure offering Psychiatric Assistance?
Dr. Elena Rossi	3.2	Generate a list of all municipalities that have more than 5 pharmacies.
Dr. Elena Rossi	3.3	What types of hospitals are present in comparison between different municipalities?
Luca Verdi	4.1	What is the total number of beds for elderly care in a municipality?
Luca Verdi	4.2	What is the total capacity of all semi-residential structures in the province?
Luca Verdi	4.3	How many health structures are managed directly by the health service in a specific municipality?

3.1.4 Concepts Identification

This section outlines the key concepts identified from the data sources and user requirements, categorizing them into Entity Types (Etypes) and defining their attributes and roles within the Knowledge Graph.

Etypes and Attributes

The identified entities are classified into three categories:

- **Core:** Central entities that directly represent the primary domain of interest (health facilities).
- **Common:** Entities that provide general context or serve as containers for other entities (geography, administrative bodies).

- **Contextual:** Entities that provide specific descriptive details or classifications for the core entities.

Table 2: *Concept Identification: Etypes and Attributes*

Entity Type	Category	Role	Attributes
HealthFacility	Core	Represents the central concept of any physical location providing healthcare services.	StructureCode (ID), Name, Address, CAP, Phone, Email, Website, Lat/Long, ServiceName
Hospital	Core	Represents a specialized facility for comprehensive medical care.	Inherits HealthFacility + ASL_Code, HospitalTypeCode, ManagementType
Dispensary	Core	Represents a facility for dispensing medication.	IVA (VAT), DispensaryType (Pharmacy/Parapharmacy)
CareStructure	Core	Represents a facility for residential or semi-residential care.	NumBeds, NumUsers, NumAdmissions, CareType, ManagementType
RehabStructure	Core	Specialized facility for therapeutic and rehabilitation services.	ASL_Code
Municipality	Common	Represents the administrative territory.	MunicipalityCode (ID), Name, Province

Relationships

The semantic relationships define how the identified entities interact to form the Knowledge Graph structure. These relationships are categorized into inheritance (class hierarchy) and semantic links (object properties).

Inheritance

These relationships define the specialization hierarchy, where specific facility types inherit attributes from the central HealthFacility class.

- **Hospital** → subClassOf → **HealthFacility**
- **Dispensary** → subClassOf → **HealthFacility**
- **CareStructure** → subClassOf → **HealthFacility**
- **RehabilitationStructure** → subClassOf → **HealthFacility**

Semantic Links (Object Properties)

These relationships connect the core entities to the common and contextual entities, establishing the domain context.

- **HealthFacility** → locatedIn → **Municipality**: Links a facility to its geographical territory.

3.1.5 ER Model

The ER model provides a structured foundation for implementing the KG, defining core entities and relationships based on the Competency Questions and the provided datasets. Our model is designed around a superclass entity (*HealthFacility*), which holds all common territorial data (address, coordinates, etc.), with specialized sub-entities inheriting from it to add specific attributes.

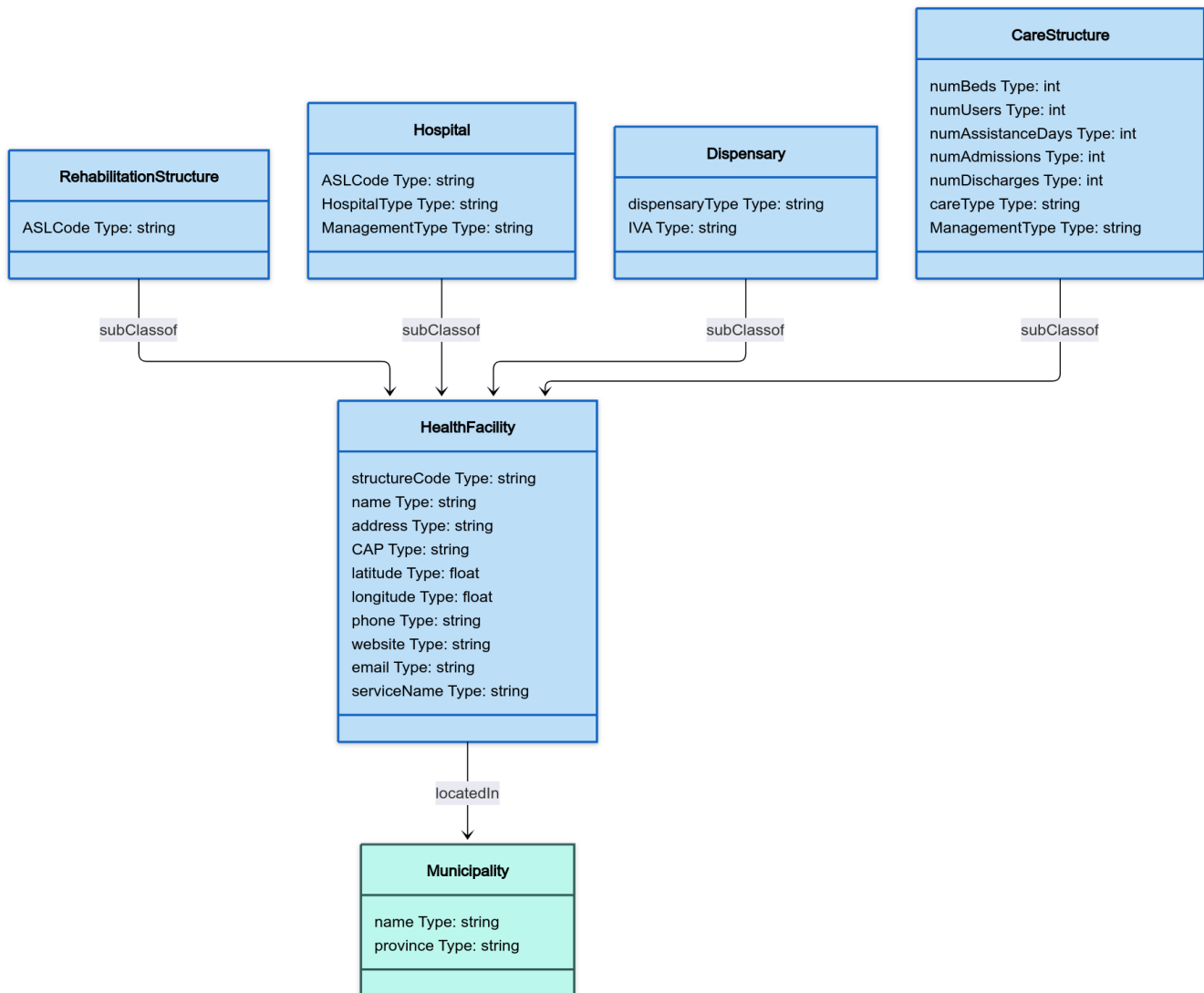


Figure 1: Entity Relation Diagram

3.2 Information Gathering

In this section, an overview of the primary input data sources available for the project is provided. This includes a list of resources that span across languages, schemas, and data values.

3.2.1 Knowledge Layer

This section outlines the semantic resources identified and utilized to structure the Knowledge Graph. For this project, Schema.org was selected as the exclusive reference ontology. This decision was driven by the need for a lightweight, highly interoperable, and web-standard vocabulary that could sufficiently model both the healthcare and territorial aspects of the domain without the excessive complexity of clinical-specific standards like FHIR.

Schema.org Vocabulary provided the foundational class hierarchy and property definitions for the majority of the entities in the Knowledge Graph. Its broad coverage allowed for a consistent modeling approach across different domains:

- **Healthcare Facilities:** The *schema : MedicalOrganization* class served as the semantic backbone for the core *HealthFacility* entity. Specific facility types were mapped directly to their Schema.org equivalents, such as *schema : Hospital* for hospitals and *schema : Pharmacy* for dispensaries.
- **Territory and Location:** The *schema : AdministrativeArea* class was used to model municipalities, providing a standard way to represent administrative boundaries. Furthermore, the *schema : Place* and *schema : GeoCoordinates* (using latitude and longitude properties) were utilized to model the physical location of every facility, ensuring compatibility with standard geospatial querying tools.
- **Organization and Attributes:** Common attributes found in the source datasets, such as addresses, tax IDs, and contact information, were mapped to standard Schema.org properties like *schema : address*, *schema : vatID*, and *schema : telephone*.

3.2.2 Data Layer

Hospital Resource The Hospital Resource is a data value dataset that provides data on hospital facilities, specifically focusing on those within the Trentino region. This resource includes detailed administrative and location metrics for each facility, offering insights into the type of hospital management and contact information. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_OSP (int64):** The unique administrative code for the hospital.
- **OSPEDALE (string):** The official name of the hospital facility.
- **INDIRIZZO (string):** The street address of the facility.
- **COMUNE (string):** The municipality where the hospital is located.

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- **TIPO_OSP (string):** Description of the hospital type (e.g., "OSPEDALE A GESTIONE DIRETTA").
 - **TELEFONO (int64):** Contact phone number.
 - **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates of the facility.

Pharmacy Resource The Pharmacy Resource is a data value dataset that provides information on pharmacies, specifically focusing on establishments within the Trentino region. This resource identifies pharmacies by their administrative codes and provides their precise location and operational start dates. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_FARMACIA (int64):** The unique administrative code for the pharmacy.
- **FARMACIA (string):** The name of the pharmacy.
- **TIPOLOGIA (string):** The classification of the pharmacy (e.g., "ORDINARIA").
- **DATA_INIZIO (string):** The date when the pharmacy commenced operations.
- **INDIRIZZO (string):** The street address.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Para-pharmacy Resource The Para-pharmacy Resource is a data value dataset that provides information on para-pharmacies (facilities authorized to sell non-prescription drugs), specifically focusing on those within the Trentino region. Similar to the Pharmacy resource, it provides location and identity data essential for mapping the pharmaceutical network. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_PARAFARMACIA (int64):** The unique code for the para-pharmacy.
- **PARAFARMACIA (string):** The commercial name of the para-pharmacy.
- **PARTITA_IVA (int64):** The VAT identification number.
- **INDIRIZZO (string):** The physical location.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Health Structures Resource The Health Structures Resource is a data value dataset that provides detailed information on general health facilities, such as polyclinics, laboratories, and outpatient centers, within the Trentino region. This dataset is particularly rich in classification data, describing the type of assistance provided and the structure's operational model. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_STRUTTURA (int64):** The unique structure identifier.

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- **STRUTTURA (string)**: The name of the health facility.
 - **TIPO_STRUTTURA (string)**: The structural classification (e.g., "AMBULATORIO E LABORATORIO").
 - **ASSISTENZA (string)**: The type of assistance provided (e.g., "ATTIVITA' CLINICA", "DIAGNOSTICA").
 - **SITO_WEB (string)**: The website of the structure.
 - **LATITUDINE_P / LONGITUDINE_P (float)**: The geographic coordinates.

Rehabilitation Structures Resource The Rehabilitation Structures Resource is a data value dataset that provides information about rehabilitation centers, specifically focusing on therapeutic communities and similar facilities in Trentino. While smaller in volume compared to other resources, it is critical for mapping specialized care. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_RIA (int64)**: The unique code for the rehabilitation structure.
- **STRUTTURA (string)**: The name of the rehabilitation facility.
- **TELEFONO (int64)**: Contact number for the facility.
- **LATITUDINE_P / LONGITUDINE_P (float)**: The geographic coordinates.

Residential Assistance Resource The Residential Assistance Resource is a data value dataset that provides utilization and capacity data for residential facilities (e.g., RSA - nursing homes), specifically focusing on assistance to the elderly and vulnerable groups in Trentino. Unlike the registry datasets, this resource contains dynamic metrics regarding usage. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **STRUTTURA (string)**: The name of the residential facility.
- **ASSISTENZA (string)**: The type of assistance provided (e.g., "ASSISTENZA AGLI ANZIANI").
- **ANNO (int64)**: The reference year for the data.
- **NUM_UTENTI (int64)**: The total number of users served.
- **NUM_POSTI (int64)**: The capacity (number of beds/places) of the structure.
- **NUM_GG_ASSISTENZA (int64)**: The total days of assistance provided.

Semi-residential Assistance Resource The Semi-residential Assistance Resource is a data value dataset that provides information about day centers and semi-residential facilities. It mirrors the structure of the Residential Assistance resource but focuses on facilities where patients do not reside permanently. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **STRUTTURA (string):** The name of the semi-residential facility.
- **TIPO_STRUTTURA (string):** The type of structure (e.g., "STRUTTURA SEMIRESIDENZIALE").
- **NUM_UTENTI (int64):** The number of users served.
- **NUM_GIORNATE (int64):** The total number of assistance days provided.

3.3 Report

3.3.1 Personas and Scenarios

We maintained the same core scenarios and personas throughout the process, refining them to better match the specific data attributes found (e.g., focusing on specific services like "Psychiatric Assistance" because that data was available).

3.3.2 Competency Questions (CQs)

Initially, our project used a set of preliminary scratch competency questions that were based on an imagined or ideal concept of the data we might have, rather than being drawn from actual information.

Once the final datasets were selected and confirmed for the project, we conducted a crucial review to ensure these questions matched the data's real content and limitations.

Adjustment: Questions were either reworded or entirely removed to be directly relevant and answerable using the chosen data.

1.
 - **Old:** "Which municipalities in the Val di Fassa are more than a 30-minute drive from the nearest Type 1 Emergency Room?"
 - **New (Dr. Elena Rossi 3.1):** "Which municipalities do not have any health structure offering Psychiatric Assistance?"
 - **Change:** The specific focus on "Val di Fassa" and "30-minute drive" was removed because the datasets did not support real-time travel time calculations. The question was broadened to a province-wide gap analysis based on available service attributes (ASSISTENZA).
2.
 - **Old:** "What specialized services (e.g., pediatrics, cardiology) are available at the primary hospital serving the remote communities of Val di Fassa?"
 - **New (Giulia Verdi 2.3):** "What specialized services are available at the health structures in a specific municipality?"

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- **Change:** The question was generalized to query any municipality, not just Val di Fassa, and simplified to match the ASSISTENZA column in the SANSTRUT001 dataset rather than inferring a "primary hospital" relationship which wasn't explicitly defined.
3.
 - **Old:** "Which local clinics (Poliambulatori) within 10km of Riva del Garda are currently accepting new primary care patients?"
 - **New (Marco Bianchi 1.3):** "List all pharmacies within a 5km radius of a specific health facility."
 - **Change:** The original question was dropped because the data sources provided no information on "accepting new patients" (real-time status). It was replaced with a feasible proximity query for pharmacies, which is supported by the coordinate data.
 4.
 - **Old:** "What is the closest Trauma Center to the road intersection at Passo Pordoi, and what is the calculated ETA via helicopter transport?"
 - **New (Marco Bianchi 1.1):** "What is the nearest hospital to a current GPS coordinates?"
 - **Change:** Calculation of "ETA via helicopter" was impossible with the static facility data. The question was simplified to a direct geodesic distance calculation ("nearest hospital") using the LATITUDINE/LONGITUDINE fields.
 5.
 - **Old:** "Which nearby hospitals are confirmed to have a functional medical evacuation helicopter pad and an available surgical team right now?"
 - **New (Luca Verdi 4.3):** "How many health structures are managed directly by the health service in a specific municipality?"
 - **Change:** Real-time resource availability ("available surgical team right now") is not in the Open Data. The question was replaced with a structural/administrative query relevant to the Planner persona (Luca Verdi).
 6.
 - **Old:** "Which Dialysis Centers within 50km of San Martino di Castrozza offer short-term appointment slots for visiting patients?"
 - **New (Luca Verdi 4.1):** "What is the total number of beds for elderly care in a municipality?"
 - **Change:** "Appointment slots" implies a booking system, which doesn't exist in the data. The question was shifted to querying capacity (NUM_POSTI), which is a hard metric provided in the Residential Care dataset.

This focused process yielded an improved and refined set of competency questions that are now more exact, useful, and clearly supported by the information in our final data sources.

3.3.3 Entity Relationship Diagrams

During the information gathering process, the ER diagram was adjusted from a previously extremely normalized manner to a general sense fitting more onto the project itself. Main changes were integrating a "Service" entity into the "HealthFacility" entity due to its flexibility, and integrating "Management" entity into the "Hospital" entity. Also we changed the Attribute